

Chapter 11, Problem 3P

Problem

A load consists of a $60\text{-}\Omega$ resistor in parallel with a $90\text{-}\mu\text{F}$ capacitor. If the load is connected to a voltage source $v_s(t) = 160 \cos 2000t$, find the average power delivered to the load.

Step-by-step solution

Step 1 of 4

Consider the voltage source $v(t)$.

$$v(t) = 160 \cos 2000t$$

Convert $v_s(t)$ into its phasor form.

$$\mathbf{V} = 160 \angle 0^\circ \text{ V}$$

Calculate the impedance of the capacitor.

$$Z_c = \frac{1}{j\omega C}$$

Substitute 2000 for ω , and 90×10^{-6} for C .

$$\begin{aligned} Z_c &= \frac{1}{j(2000)(90 \times 10^{-6})} \\ &= -j5.55 \text{ }\Omega \end{aligned}$$

Step 2 of 4

Draw the circuit as shown in figure 1.

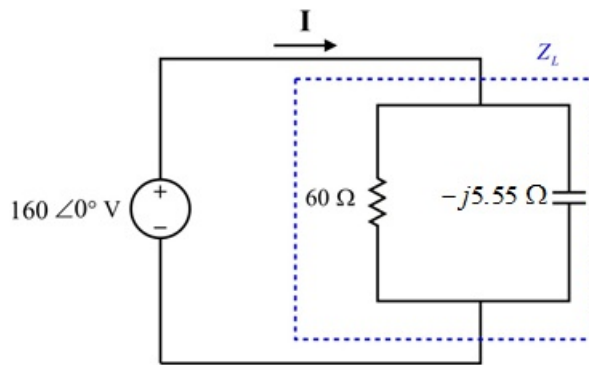


Figure 1

Step 3 of 4

Calculate the impedance of load Z_L shown in figure 1.

$$\begin{aligned} Z_L &= (60) \parallel (-j5.55) \\ &= \frac{(60)(-j5.55)}{60 - j5.55} \\ &= \frac{-j333.33}{60 - j5.55} \\ &= 0.51 - j5.51 \, \Omega \end{aligned}$$

Calculate the current through the impedance Z_L .

$$\mathbf{I} = \frac{\mathbf{V}}{Z_L}$$

Substitute $160\angle 0^\circ \text{ V}$ for $\mathbf{V}$, and $0.51 - j5.51 \, \Omega$ for Z_L .

$$\begin{aligned} \mathbf{I} &= \frac{160\angle 0^\circ}{0.51 - j5.51 \, \Omega} \\ &= \frac{160}{5.532\angle -84.715^\circ} \\ &= 28.92\angle 84.715^\circ \text{ A} \end{aligned}$$

Step 4 of 4

The current through the impedance Z_L in time-domain is,

$$i(t) = 28.92 \cos(2000t + 84.715^\circ) \text{ A}$$

Calculate the average power delivered to the load Z_L .

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

Substitute 160 for V_m , 28.92 for I_m , 0° for θ_v , and 84.715° for θ_i .

$$\begin{aligned} P &= \frac{1}{2} (160)(28.92) \cos(0^\circ - 84.715^\circ) \\ &= (2313.81) \cos(-84.715^\circ) \\ &= (2313.81)(0.092) \\ &= 213.12 \text{ W} \end{aligned}$$

Therefore, the average power delivered to the load is 213.12 W.